

Q3: (a)

Private: 10.0.0.0/24

Public: 203.0.113.50

- Host to internet: Create a table entry by replacing packet's source IP & port and enter it inside table.
- Internet to host: Match destination IP & port from table and replace IPs. Then, forward it to ~~the~~ internal network.

(b)

Private IP / Port	Public IP / Port Port
10.0.0.5 / 57	203.0.113.50 / 6000
10.0.0.6 / 57	203.0.113.50 / 6003
10.0.0.7 / 57	203.0.113.50 / 7000

(c)

In full-clone NAT, once a mapping is created, then any external host can send packets to the public IP & NAT will forward to the internal host.

i.

Match	Action
<u>Outbound</u>	
Match on source IP (private), source port, protocol.	Rewrite source IP to public IP, source port to assigned public port, then forward.
<u>Inbound</u>	
Match on destination IP (public), destination port, protocol.	Rewrite destination IP to private IP, destination port to private port, then forward.

ii.

Yes, it maintains a flow table entry for active mapping. For any incoming packets it just rewrites IP ~~with~~ to internal host IP since multiple hosts can send.